

Desiree Y. Phua
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Education

- 2016 **Doctor of Philosophy**
Nanyang Technological University, Singapore
The complexity of individual and the social-cultural ecology: Interaction of genes, culture, and attachments.
- 2009 **B.A. in Psychology (2nd Class Honors, Upper Division)**
with minor in Sociology
School of Humanities & Social Sciences, Nanyang Technological University

Professional Appointments

- 2023 – **Research Scientist**
present Translational Neurosciences
Institute for Human Development & Potential, A*STAR
- 2024 – **Project Team Member**
2026 “Social Determinants of Mental Wellbeing in Young People” project
International Science Council & World Health Organization
- 2020 – 2023 **Senior Research Fellow**
Translational Neurosciences
Singapore Institute for Clinical Sciences, A*STAR
- 2016 – 2020 **Research Fellow**
Translational Neurosciences
Singapore Institute for Clinical Sciences, A*STAR
- 2015 – 2016 **Research Associate**
Nanyang Business School
Nanyang Technological University, Singapore

Grants

- Apr 2024 – **Open-Fund Young Individual Research Grant, S\$322,595.00**
Apr 2027 **Principal Investigator**
Maximizing the effectiveness of public health messaging to improve adolescent mental health in Singapore
MOH National Medical Research Council, Singapore
- Oct 2024 – **Population Health Research Grant, S\$1,700,933.00**
Oct 2029 **Collaborator**
Adolescents’ health risk behaviours in the Singapore context – perspectives, participatory action and predictors (PI: Dr. Mary Chong, NUS)
MOH National Medical Research Council, Singapore
- Dec 2021 – **Social Sciences Innovation Seed Fund, S\$350,990.00**
Dec 2023 **Principal Investigator**
Adolescents’ positive mental health: Its operationalization and correlates

*Social Sciences & Technology Horizontal Technology Programme Office
Agency for Science, Technology, and Research*

Publications

- Phua, D.Y.**, Broekman, B.F.P., Saw, S-M., Gluckman, P.D., Erikson, J.G., & Rodriguez, A. (under review). The evolving architecture of adolescent resilience traits across sociocultural and developmental contexts.
- Madan, S., Savani, K., Mehta, P.H., **Phua, D.Y.**, ..., & Morris, M.W. (2025). Stress reactivity and sociocultural learning: More stress-reactive individuals are quicker at learning sociocultural norms from experiential feedback. *Journal of Personality and Social Psychology*, 128(6), 1292–1314. <https://doi.org/10.1037/pspi0000487>
- Phua, D.Y.**, & Chew, C.S.M. (2025). Core features of positive mental health in adolescents and their protective role against psychopathology. *Scientific Reports*, 15, 4228. <https://doi.org/10.1038/s41598-025-88454-x>
- Yeo S.L., **Phua, D.Y.**, & Hong, Y. (2025). The effects of dangerous world beliefs on COVID-19 preventive behaviors in Singapore: The moderating role of public health communication. In R. T. Juan Meng (Ed.), *Strategic Communication and the Global Pandemic* (pp. 137-150). Routledge. <https://doi.org/10.4324/9781003466437>
- Phua, D.Y.**, Chen, H., Chong, Y.S., ..., & Meaney, M.J. (2023). Allostatic load in children: The cost of empathic concern. *Proceedings of the National Academy of Sciences*, 120(39), e2217769120. <https://doi.org/10.1073/pnas.2217769120>
- Phua, D.Y.**, Chew, C.S.M., Tan, Y.L., Ng, B.J.K., Lee, F.K.L., & Tham, M.M.Y. (2023). Differential effects of prenatal psychological distress and positive mental health on offspring socioemotional development from infancy to adolescence: A meta-analysis. *Frontiers in Pediatrics*, 11, 1221232. <https://doi.org/10.3389/fped.2023.1221232>
- Anant, N., Kaur, D., Nadarajan, R., **Phua, D.Y.**, Chong, Y.S., ..., & Ang, Y.S. (2023). Validating the Children's Depression Inventory-2: Results from the Growing Up in Singapore Towards Healthy Outcomes (GUSTO) study. *PLOS ONE*, 18(5), e0286197. <https://doi.org/10.1371/journal.pone.0286197>
- Cai, S., **Phua, D.Y.**, Tham, E.K., Goh, D.Y.T., Teoh, O.H., Shek, L.P.C., ..., & Meaney, M.J. (2023). Mid-pregnancy and postpartum maternal mental health and infant sleep in the first year of life. *Journal of Sleep Research*, 32(3), e13804. <https://doi.org/10.1111/jsr.13804>
- Yeo, S.L., **Phua, D.Y.**, & Hong, Y. (2022). The effects of lay beliefs on disease preventive behaviors during Covid-19 pandemic in Singapore: The moderating role of public health communication. *International Journal of Strategic Communications*, 16(3), 485-498. <https://doi.org/10.1080/1553118X.2022.2036742>
- Aishworiya, R., Magiati, I., **Phua, D.Y.**, ..., & Law, E. (2022). Are there bidirectional influences between screen time exposure and social behavioral traits in young children? *Journal of Developmental & Behavioral Pediatrics*. <https://doi.org/10.1097/DBP.0000000000001069>
- Sum, K. K., Tint, M. T., Aguilera, R., Dickens, B. S. L., Choo, S., Ang, L. T., **Phua, D.Y.**, ..., &

- Huang, J. (2022). The socioeconomic landscape of the exposome during pregnancy. *Environment International*, 163, 107205. <https://doi.org/10.1016/j.envint.2022.107205>
- Kee, M.Z.L., Ponmudi, S., **Phua, D.Y.**, ..., & Meaney, M.J. (2021). Preconception origins of perinatal maternal mental health. *Archives of Women's Mental Health*, 24(4), 605-618. <https://doi.org/10.1007/s00737-020-01096-y>
- Syn, N.L, Chan, S-Y., Chia, E.W.Y, Ong, W.X., **Phua, D.Y.**, Cai, S., ..., & Law, E.C. (2021). Severity of nausea and vomiting in pregnancy and early childhood neurobehavioural outcomes: The Growing Up in Singapore Towards Healthy Outcomes study. *Paediatric and Perinatal Epidemiology*, 35(1), 98-108. <https://doi.org/10.1111/ppe.12703>
- Phua, D.Y.**, Chen, H., Chong, Y.S., Gluckman, P.D., Broekman, B.F.P., & Meaney, M.J. (2020). Network analyses of maternal pre- and post-partum symptoms of depression and anxiety. *Frontiers in Psychiatry*, 11, 785. <https://doi.org/10.3389/fpsy.2020.00785>
- Phua, D.Y.**, Leong, C.H., & Hong, Y. (2020). Heterogeneity in national identity construct: Example of Singapore using network analysis. *International Journal of Intercultural Relations*, 78, 20-32. <https://doi.org/10.1016/j.ijintrel.2020.05.010>
- Phua, D.Y.**, Kee, M.Z.L., & Meaney, M.J. (2020). Positive maternal mental health, parenting, and child development. *Biological Psychiatry*, 87(4), 328-337. <https://doi.org/10.1016/j.biopsych.2019.09.028>
- Loo, E., Soh, S.E., ..., **Phua, D.Y.**, ..., Cheng, Z.R. (2020). Cohort profile: Singapore preconception study of long-term maternal and child outcomes (S-PRESTO). *European Journal of Epidemiology*. <https://doi.org/10.1007/s10654-020-00697-2>
- van Lee, L., Chia, A., **Phua, D.Y.**, ..., Chong, M. F. F. (2020). Multiple modifiable lifestyle factors and the risk of perinatal depression during pregnancy: Findings from the GUSTO cohort. *Comprehensive Psychiatry*. <https://doi.org/10.1016/j.comppsy.2020.152210>
- Aishworiya, R., Cai, S., Chen, H.Y., **Phua, D.Y.**, Broekman, F.P., ..., & Law, E.C. (2019). Television viewing and child cognition in a longitudinal birth cohort in Singapore: The role of maternal factors. *BMC Pediatrics*, 19(10), 286. <https://doi.org/10.1186/s12887-019-1651-z>
- Phua, D.Y.**, Kee, M.Z.L., Koh, D.X.P., Rifkin-Graboi, A., Daniel, M., Chen, H., ..., & Meaney, M.J. (2017). Positive maternal mental health during pregnancy associated with specific forms of adaptive development in early childhood: Evidence from a longitudinal study. *Development & Psychopathology*, 29(5), 1573-1587. <https://doi.org/10.1017/S0954579417001249>
- Phua, D.Y.**, Meaney, M.J, Khor, C. C., Lau, I.Y. & Hong, Y. (2017). Effects of bonding with parents and home culture on intercultural adaptations and the moderating role of genes. *Brain Behavioral Research*, 325, 223-236. <https://doi.org/10.1016/j.bbr.2017.02.012>
- Phua, D.Y.**, & Christopoulos, G. (2014). Social neuroscience tasks: Employing fMRI to understand the social mind. In T.D. Papageorgiou, G.I. Christopoulos & S.M. Smirnakis (Eds), *Advanced Brain Neuroimaging Topics in Health and Disease - Methods and Applications* (chap 25). <https://doi.org/10.5772/58279>

- Hong, Y., Fang, Y., Yang, Y., & **Phua, D.Y.** (2013). Cultural attachment: A new theory and method to understand cross-cultural competence. *Journal of Cross-Cultural Psychology*, 44(6), 1024-1044. <https://doi.org/10.1177/0022022113480039>
- Hong, Y., & **Phua, D.Y.** (2013). In search of culture's role in influencing individual social behaviour. *Asian Journal of Social Psychology*, 16(1), 26-29. <https://doi.org/10.1111/ajsp.12016>
- Phua, D.Y.**, Rifkin-Graboi, A., Saw, S.M., Meaney, M.J., Qiu, A. (2012). Executive functions of six-year-old boys with normal birth weight and gestational age. *PLoS ONE*, 7(4): e36502. <https://doi.org/10.1371/journal.pone.0036502>
- Qiu, A., Rifkin-Graboi, A., Zhong, J., **Phua, D.Y.**, Meaney, M.J. (2011). Birth weight and gestation influence striatal morphology and motor response in six-year-old boys. *NeuroImage*, 59(2), 1065-70. <https://doi.org/10.1016/j.neuroimage.2011.09.032>
- Zhong, J., **Phua, D.Y.**, & Qiu, A. (2010). Quantitative evaluation of LDDMM, FreeSurfer, and CARET for cortical surface mapping. *NeuroImage*, 52(1), 131-141. <https://doi.org/10.1016/j.neuroimage.2010.03.085>

Conference Presentations

- Phua, D.Y.** & Wong, L.Z. (2024). *Network analysis of adolescent positive mental health*. Poster presented at 2024 Association of Psychological Science Annual Convention, San Francisco, USA.
- Phua, D.Y.**, Chew, C. Meaney, M.J. (2023). *Clash of worldviews: Moderating effect of implicit theory of social status on growth mindset and adolescents' well-being*. Poster presented at 2023 International Convention of Psychological Science (ICPS), Brussels, Belgium.
- Cai, S., **Phua, D.Y.**, Tham, E.K.H., ..., Meaney, M.J. (2019). *Maternal mood & infant sleep in the first year of life*. Paper presented at 11th World Congress on Development Origins of Health and Disease, Melbourne, Australia.
- Koh, D.X.P., **Phua, D.Y.**, Qiu, A., Birit, B.F.P., Daniel, M.L., Chen, H., Chong, Y.S., Gluckman, P., & Meaney, M.J. (2019). *Prenatal positive maternal mental health associates with inter-individual variation neonatal amygdala structure*. Poster presented at International Convention of Psychological Science, Paris, France.
- Van Lee, L., Chia, A., **Phua, D.Y.**, Colega, M., ..., Chong, M.F. (2019) *Multiple modifiable lifestyle factors and the risk of perinatal depression during pregnancy: The GUSTO cohort*. Poster presented at 13th European Nutrition Conference, Dublin, Ireland.
- Phua, D. Y.**, Kee, M.Z.L., Koh, D.X.P., ..., Meaney, M.J. (2017). *Positive maternal mental health during pregnancy associated with specific forms of adaptive development in early childhood*. Paper presented at 10th World Congress on Development Origins of Health and Disease, Rotterdam, The Netherlands.
- Phua, D.Y.**, Hong, Y., & Leong, C.H. (2017). *Disagreement about who can be part of us: An examination of heterogeneity of the social markers in Singapore using exploratory factor mixture model*. Symposium at 10th Biennial Conference for the International Academy for Intercultural Research, New York, USA.
- Phua, D.Y.**, Meaney, M.J., & Hong, Y. (2016). *Parents, a boon or bane? Differential effects of parenting in acculturation*. Symposium at 23rd International Congress of the International

Association for Cross-Cultural Psychology, Nagoya, Japan.

- Phua, D.Y.,** Meaney, M.J., & Hong, Y. (2016). *Differential effects of childhood parental attachment on acculturation in adulthood*. Poster presented at 17th Society for Personality & Social Psychology Annual Convention, San Diego, USA.
- Low, B., **Phua, D.Y.,** ..., Liu, J.C.Y. (2015). *Variation in OXTR genotype is associated with early childhood adiposity*. Paper presented at 45th Annual Meeting of Society for Neuroscience, Chicago, USA.
- Phua, D.Y.,** Meaney, M.J., & Hong, Y. (2015). *Moving across cultures: Feeling overprotected by father & attachment to home mattered*. Poster presented at International Convention for Psychological Science, Amsterdam, Netherlands.
- Phua, D.Y.,** Meaney, M.J., & Hong, Y. (2014). *Moderating effect of genetic on relationship with parents, worldview & experience of studying abroad*. Poster presented at Society for Social Neuroscience conference, Washington D.C., USA.
- Phua, D.Y.,** Meaney, M.J., Holbrook, J.D., & Hong, Y. (2013). *Genes and lay beliefs play a role in the experiences of studying aboard: Moderating effects of 5-HTTLPR polymorphism*. Poster presented at International Cultural Neuroscience Consortium (ICNC) conference, Evanston, USA
- Phua, D.Y.,** Kim, J., Liu, Y. & Shan, W. (2013) *Warring identities: Management of multiple professional identities among creative entrepreneurs*. Paper presented at Asian Perspectives on Creative Industries conference, Singapore.
- Phua, D.Y.,** Zhan, S., Ang, C., & Hong, Y. (2013). *What is the Singaporean identity, and why or how does it matter?* Paper presented at 7th Self Biennial International Conference, Singapore.

Invited Talks

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| 2024 | <i>Positive mental health in adolescents & their protective role against psychopathology</i>
Ministry of Education, Singapore |
| 2022 | <i>Structure of adolescent resilience and the relationship to poor mental health</i>
Ministry of Education, Singapore |
| 2021 | <i>Maternal well-being and its effects on children</i>
Touch Family Conference, Singapore |
| 2018 | <i>Distinctiveness of positive maternal mental health</i>
KK Women's & Children's Hospital, Singapore |
| 2014 | <i>Genes and lay beliefs play a role in the experiences of studying aboard</i>
Kyoto University, Japan |
| 2013 | <i>Introduction to Social Neuroscience</i>
Division of Psychology, School of Humanities and Social Sciences, NTU |

Campus/Departmental Talks

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| 2025 | <i>Understanding youth well-being through profiles of resilience and mental health</i>
Integrated Adolescent Research Programme presentation to Ministry of Education |
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- and secondary school leaders
Institute for Human Development and Potential (IHDP, A*STAR)
- 2024 *Nomological network of adolescent psychopathology and well-being*
IHDP Research Meeting
Institute for Human Development and Potential (IHDP, A*STAR)
- 2023 *Mental Health from the Perspective of Adolescents*
SICS Mental Health Workshop
Singapore Institute for Clinical Sciences (SICS, A*STAR)
- What is positive mental health to youths? Addressing the question through different network analyses*
Social Sciences & Technology Seminar
Agency for Science, Technology, & Research (A*STAR)
- Positive mental health in adolescents*
Translational Neurosciences Seminar
Singapore Institute for Clinical Sciences (SICS, A*STAR)
- Clash of Worldviews: Moderating effect of implicit theory of social status on growth mindset & adolescents' well-being*
SICS Research Meeting
Singapore Institute for Clinical Sciences (SICS, A*STAR)
- 2021 *Cumulative risk & allostatic load in children: The cost of caring*
SICS Research Meeting
Singapore Institute for Clinical Sciences (SICS, A*STAR)
- 2015 *Culture & Sociogenomics: Culture, Psychology & Genetics*
Nanyang Business School, NTU

Teaching Experience

- 2024 – present *Data Literacy for Healthcare*
Adjunct Lecturer
Saw Swee Hock School of Public Health, National University of Singapore
- 2023 *Psychological and Sociological Perspectives for Health Sciences*
Associate Faculty
Health and Social Sciences cluster, Singapore Institute of Technology
- 2016 *Workshop on Latent Class Analysis*
Instructor
Institute of Policy Studies, National University of Singapore
- 2013 *Organizational Behavior & Design*
Instructor
Nanyang Business School, Nanyang Technological University

Other Certificates

- 2019 **Introduction to Behavioral Insights**
Temasek Polytechnic
- 2015 **Post-Diploma in Cell & Molecular Biology**
Ngee Ann Polytechnic, Singapore
- 2015 **Certificate in University Teaching for Teaching Assistants**
Nanyang Technological University

Academic Service

- Reviewer Journal of Research on Adolescence
Journal of Affective Disorders
Transcultural Psychiatry
Frontiers in Psychology
Journal of Personality and Social Psychology
Asian Journal of Social Psychology
Social and Personality Psychology Compass

Community Involvement/Outreach

- 2025 *Communicating scientific research*
Course for secondary one students
School of Science and Technology
- 2024 *The amazing adolescent brain*
Talk for secondary two students
Orchid Park Secondary School
- Mentor for Science Competition (Science Castle Singapore)*
Mentor a group of secondary two students on a psychology project
School of Science and Technology
- 2023 *Youth positive mental health*
Research project with School of Science and Technology
- 2020 *Mental health of secondary one students*
Research consultancy project with Northbrooks Secondary School
Scientist-In-School program by A*STAR & Ministry of Education
- 2019 *Raising awareness of positive mental health*
One-North Festival
Agency for Science, Technology & Research, Singapore (A*STAR)

Awards and Honors

- 2016 SPSP Travel Award
Society for Personality & Social Psychology
- 2013 ICNC Student Travel Award
International Cultural Neuroscience Consortium
- 2011 – 2015 Research Scholarship

Ministry of Education, Singapore

Courses Prepared to Teach

Health Communication
Adolescence in the Digital World of the 21st Century
Social Determinants of Health and Well-Being
The Pursuit of Well-Being and Happiness
Social Psychology
Fundamentals of Social Science Research

References

Michael J. Meaney

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